# QD-SEARCH

*Quality-Diversity Algorithms for Autonomous AI Research*

UCSB NLP Lab

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## THE PROBLEM

Why diversity matters

Standard LLM-driven research converges to dominant solutions. Agents find one good approach and exploit it — missing diverse high-quality alternatives.

**Feb 2026 study:**
Only 2/5 tasks showed genuine diversity without explicit QD pressure

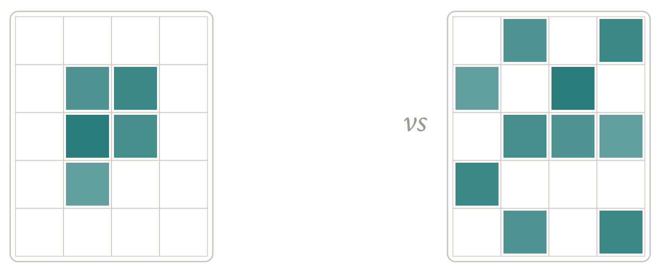
### Quality-Diversity Algorithms

QD algorithms optimize for two objectives at once: solution quality (fitness) and behavioral diversity (coverage). Instead of returning a single best answer, they maintain an archive of high-performing solutions across different regions of the design space.

**MAP-Elites (example):**
Divides the feature space into a grid of cells. Each cell stores only the best solution found for that niche. New solutions compete to enter the archive — improving quality while filling the map.

**Greedy vs. MAP-Elites**

Each cell = a different approach niche



Greedy Search — 5 cells / all clustered

*vs*

MAP-Elites — 10 cells / spread out

**Diversity is a hidden objective**

*in ML research — easy to miss, hard to recover*

## THE APPROACH

Quality-Diversity Search Loop

Three fixed stages form the research harness. The QD algorithm is a composable, pluggable layer that decides what to explore next.

HARNESS

- **IDEATE** — LLM proposes strategies
- **EXECUTE** — Parallel bwrap sandboxes
- **EVALUATE** — Score + extract features

IDEATE → EXECUTE → EVALUATE

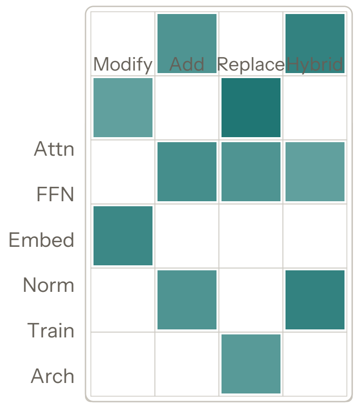
results ↓ · ↑ parents + context

### QD ALGORITHM LAYER

*composable · pluggable*

**Feature Archive**

Each cell = one approach niche.
Only the best solution per cell is kept.




11 / 24 cells filled

**Composable Algorithms**

Swap any algorithm into the loop:

- `MAP-Elites` — Keep the best solution in each grid cell
- `CVT-MAP-Elites` — Adaptive Voronoi regions instead of fixed grid
- `CMA-ME` — Learn which search directions improve the archive
- `Novelty Search` — Reward being different, regardless of fitness
- `Curiosity-Driven` — Prioritize the most uncertain regions
- `Go-Explore` — Random exploration first, then optimize

### Key Innovations

- **LLM feature classifier** — 81% accuracy via Gemini
- **Parallel bwrap sandboxes** — GPU isolation per executor
- **Cell-targeted exploration** — Bias toward empty niches
- **Persistent ideator memory** — Context carries across turns
- **Agent-agnostic harness** — OpenCode · Claude · Cursor · Codex
- **Pluggable search strategy** — Greedy, MAP-Elites, or custom

| 23% | 10.5 | 0.951 |
|---|---|---|
| Coverage | QD Score | Best Fitness |
| niches explored | aggregate fitness | val_bpb (lower=better) |

## RESULTS & FUTURE

Evidence + Roadmap

**NanoGPT Optimization (100 iterations)**

| Best BPB | Coverage |
|---|---|
| **0.9507** | **23%** |
| 3.3% below baseline | 16 of 70 niches filled |

**Discovered approaches:**

- Grouped Query Attention (GQA)
- SwiGLU gating mechanisms
- Deeper models (12–16 layers)
- Novel learning rate schedules
- Embedding modifications

### ALGORITHM ROADMAP

**Implemented**

- `MAP-Elites (Grid)` — Fixed bins — keeps the best solution found in each cell of a feature grid.
- `CVT-MAP-Elites` — Adaptive Voronoi tessellation partitions continuous feature spaces naturally.
- `LLM Feature Classifier` — Gemini classifies solutions into archive cells with 81% accuracy.

**In Pipeline**

- `CMA-ME` — Learns which search directions in feature space produce the most archive gains.
- `Novelty Search` — Rewards behavioral novelty — solutions that differ most from everything seen so far.
- `Curiosity-Driven` — Prioritizes exploration where the model's predictions are most uncertain.
- `Novelty Checker` — Validates that proposed ideas are genuinely novel via archive embedding comparison.
- `Island Model` — Multiple independent populations that periodically exchange elite solutions.
- `Go-Explore` — Finds diverse states through random exploration, then optimizes the most promising.
- `Multi-Task Archive` — Single archive spanning domains — enabling cross-task knowledge transfer.
- `Dynamic Grids` — Auto-splits dense archive regions and merges sparse ones as coverage evolves.

**Related:** CodeEvolve (arXiv:2510.14150) · AIRA-Dojo (arXiv:2507.02554) · AIDE (arXiv:2502.13138) · OMNI-EPIC (arXiv:2405.15568)

**Core thesis:** *Explicit diversity pressure yields qualitatively different — and often superior — solutions that greedy optimization misses.*